

#W[1] = FPT: Fixed-Parameter Tractable Exact Algorithms for the #k-Matching Problem

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The concept of NP-completeness has been proposed for half a century, and it is conjectured that there are no subexponential-time algorithms for NP-hard problems, which is known as the Exponential Time Hypothesis (ETH). As a pivotal conjecture in the field of theoretical computer science, numerous conjectures in computer science rely on ETH. A corollary of the Exponential Time Hypothesis is the Counting Exponential Time Hypothesis (#ETH), and a further corollary of #ETH is that $\#W[1] \neq \text{FPT}$. The #k-matching problem is a well-known $\#W[1]$ -complete problem. We have discovered an algorithm for the #k-matching problem with a running time of $f(k)n^{O(1)}$. This result implies that the hypotheses $\#W[1] \neq \text{FPT}$, $W[1] \neq \text{FPT}$, the Counting Exponential Time Hypothesis, and the Exponential Time Hypothesis all do not hold.

CCS Concepts: • **Theory of computation** → **Complexity theory and logic**.

Additional Key Words and Phrases: Computational Complexity, Parameterized Complexity, #k-Matching, Exponential Time Hypothesis, Fixed-Parameter Tractability

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1 Introduction

Since the formal proposal of the concept of NP-completeness, the field of theoretical computer science has engaged in half a century of in-depth exploration and intense debate on the computational complexity of such problems [2]. The core challenge of NP-hard problems is that no polynomial-time algorithm has been found for their solution to date, and it is widely conjectured that there exist no efficient subexponential-time algorithms for these problems. This conjecture has been condensed into the highly influential Exponential Time Hypothesis (ETH) in theoretical computer science [5]. As a foundational conjecture in the complexity theory framework, ETH not only profoundly characterizes the computational lower bounds of NP-hard problems but also serves as the logical premise for numerous important computational conjectures and theoretical deductions. Verifying its rationality and validity has long been a research hotspot and core topic in the field.

The value of ETH lies not only in its characterization of NP-hard problems themselves but also in its ability to derive a series of far-reaching complexity conclusions, including a key corollary in the field of parameterized complexity: $\#W[1] \neq \text{FPT}$. Parameterized complexity is a refined complexity analysis framework for intractable problems, whose core is to decompose the input of a problem into two parts—the *main input* (with size denoted by n) and the *parameter* (denoted by k)—and characterize the impact of the parameter on the time complexity of an algorithm independently. A Fixed-Parameter Tractable (FPT) algorithm is the core criterion for efficient algorithms in this framework, with a

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strict definition: a problem is fixed-parameter tractable (FPT) if there exists a computable function $f(k)$ that depends only on the parameter k , such that the algorithm can solve the problem in $f(k) \cdot n^{O(1)}$ time. Here, $f(k)$ is typically an exponential or higher-order function of k , and $n^{O(1)}$ denotes polynomial time with respect to the main input size n . This complexity form implies that when the parameter k is much smaller than the input size n , the algorithm can handle a main input of any size in polynomial time of n . The relationship between $\#W[1]$ and FPT directly determines whether a class of counting problems admits parameterized efficient algorithms, serving as an important link connecting classical complexity and parameterized complexity. ETH naturally implies the Counting Exponential Time Hypothesis ($\#ETH$), and it has been proven that $\#W[1] \neq \text{FPT}$ under $\#ETH$ [4], meaning that no FPT algorithms exist for $\#W[1]$ -hard problems. In addition, $W[1] \neq \text{FPT}$ is another important conjecture in parameterized complexity [1], from which $\#W[1] \neq \text{FPT}$ can be directly deduced.

As a classic problem in counting complexity, the $\#k$ -matching problem has been proven to be $\#W[1]$ -hard. By means of an extremely concise elementary algebraic method, we present an algorithm for the $\#k$ -matching problem with a running time of $f(k)n^{O(1)}$. This result implies that the long-standing conjectures of $\#W[1] \neq \text{FPT}$, $W[1] \neq \text{FPT}$, the Counting Exponential Time Hypothesis, and the Exponential Time Hypothesis all do not hold.

2 Method

We study the k -matching counting problem. Let $G = (V, E)$ denote the host graph for which we count k -matchings, A be the adjacency matrix of G , $A_{[i,j]}$ represent the (i, j) -th entry of A , $n = |V|$ be the number of vertices in G , $N(k)$ be the number of k -matchings in G , and $P(n, k)$ be the set of k -permutations of $\{1, 2, \dots, n\}$.

The set $P(n, n)$ contains $n!$ permutations, denoted as $\phi_1, \phi_2, \dots, \phi_{n!}$, where $\phi(j)$ denotes the j -th element of the n -permutation ϕ . We denote $n!$ subgraphs of G as $G_1 = (V_1, E_1), G_2 = (V_2, E_2), \dots, G_{n!} = (V_{n!}, E_{n!})$, where for all $1 \leq i \leq n!$, $V_i = V$ and $E_i = \{(a, b) \mid (a, b) \in E \wedge b = \phi_i(a)\}$.

Since each vertex j in G_i has at most one edge $(j, \phi_i(j))$, the number of k -directed matchings in G_i can be written as $\frac{1}{k!} \sum_{\sigma \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]}$. Here, a k -directed matching is defined as a set of k directed edges passing through $2k$ distinct vertices by definition.

Any k -directed matching $\{< i_1, j_1 >, < i_2, j_2 >, \dots, < i_k, j_k >\}$ in G is a k -directed matching in G_i if and only if $\phi_i(i_h) = j_h$ for all $1 \leq h \leq k$. There are exactly $(n - k)!$ such permutations in $P(n, n)$. Meanwhile, one k -matching corresponds to 2^k k -directed matchings. Thus, we have:

Lemma 1. $N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{i=1}^{n!} \sum_{\sigma \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]}$.

Lemmas 2 and 3 are trivial:

Lemma 2. $\sum_{\sigma \in P(n, 2)} x_{\sigma(1)} x_{\sigma(2)} = \sum_{\sigma(1)=1}^n \sum_{\sigma(2)=1}^n x_{\sigma(1)} x_{\sigma(2)} - \sum_{\sigma(1)=1}^n x_{\sigma(1)} x_{\sigma(1)}$.

Lemma 3. If $\forall \leq i \leq n, x_i \in \{0, 1\}$, then $\sum_{\sigma \in P(n, 2)} x_{\sigma(1)} x_{\sigma(2)} = \sum_{\sigma(1)=1}^n \sum_{\sigma(2)=1}^n x_{\sigma(1)} x_{\sigma(2)} - \sum_{\sigma(1)=1}^n x_{\sigma(1)}$.

It is not difficult to derive from Lemma 3 that:

Lemma 4. If $\forall 1 \leq i \leq n, x_i \in \{0, 1\}$, then $\forall k > 1$, there exist $g'_k(1), g'_k(2), \dots, g'_k(k)$ such that

$$\sum_{\sigma \in P(n, k)} \prod_{i=1}^k x_{\sigma(i)} = \sum_{l=1}^k g'_k(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} = \sum_{l=1}^k g'_k(l) \prod_{h=1}^l \sum_{1 \leq i_h \leq n} x_{i_h}.$$

Proof. We prove this by mathematical induction:

Base case: Already established by Lemma 3.

Inductive step: The inductive hypothesis is

$$\sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} = \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h}. \quad (1)$$

The term $\sum_{\sigma \in P(n, k)} \prod_{i=1}^k x_{\sigma(i)}$ can be rewritten as

$$\begin{aligned} \sum_{\sigma(k)=1}^n \sum_{\substack{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1) \\ \sigma(1), \sigma(2), \dots, \sigma(k-1) \neq \sigma(k)}} \prod_{i=1}^k x_{\sigma(i)} &= \sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - \sum_{j=1}^{k-1} \sum_{\sigma \in P(n, k-1)} x_{\sigma(j)} \prod_{i=1}^{k-1} x_{\sigma(i)} \\ &= \sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - (k-1) \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)}. \end{aligned} \quad (2)$$

From the inductive hypothesis:

$$\begin{aligned} \sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} &= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \sum_{i=1}^n x_i \prod_{h=1}^l x_{i_h} \\ &= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \sum_{i_{l+1}=1}^n x_{i_{l+1}} \prod_{h=1}^l x_{i_h} \\ &= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_{l+1} \leq n} \prod_{h=1}^{l+1} x_{i_h}. \end{aligned} \quad (3)$$

Substituting (1) and (3) into (2), we obtain:

$$\begin{aligned} \sum_{\sigma \in P(n, k)} \prod_{i=1}^k x_{\sigma(i)} &= \sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - (k-1) \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} \\ &= g'_{k-1}(k-1) \sum_{1 \leq i_1, i_2, \dots, i_k \leq n} \prod_{h=1}^k x_{i_h} \\ &\quad + \sum_{l=2}^{k-1} g'_{k-1}(l-1) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} \\ &\quad - (k-1) \sum_{l=2}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} - (k-1) g'_{k-1}(1) \sum_{1 \leq i_1 \leq n} x_{i_1} \\ &= \sum_{l=2}^{k-1} (g'_{k-1}(l-1) - (k-1) g'_{k-1}(l)) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} \\ &\quad + g'_{k-1}(k-1) \sum_{1 \leq i_1, i_2, \dots, i_k \leq n} \prod_{h=1}^k x_{i_h} - (k-1) g'_{k-1}(1) \sum_{1 \leq i_1 \leq n} x_{i_1}. \end{aligned} \quad (4)$$

Let

$$g'_k(l) = \begin{cases} g'_{k-1}(k-1), & l = k, \\ g'_{k-1}(l-1) - (k-1)g'_{k-1}(l), & 1 < l < k, \\ -(k-1)g'_{k-1}(1), & l = 1. \end{cases}$$

Furthermore, let

$$g'_2(l) = 1 \quad \forall 1 \leq l \leq 2.$$

This completes the inductive step.

Clearly, we only need $O(k^3)$ time to compute $g_k(l)$ according to the recursive formula in the proof of Lemma 4.

It follows from Lemma 4 that:

Theorem 1. $N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{i=1}^{n!} \prod_{h=1}^l A_{[j_h, \phi_i(j_h)]}$.

Proof. From Lemma 1:

$$N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{i=1}^{n!} \sum_{\sigma \in P(n,k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]} \quad (5)$$

From Lemma 4:

$$\sum_{\sigma \in P(n,k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]} = \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l A_{[j_h, \phi_i(j_h)]}. \quad (6)$$

Substituting (6) into (5), we obtain:

$$N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{i=1}^{n!} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l A_{[j_h, \phi_i(j_h)]}. \quad (7)$$

Interchanging the order of summation yields Theorem 1.

It is straightforward to derive Theorem 2 from Theorem 1:

Theorem 2. $N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k (n-l)! g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}$.

Proof. For $\forall \phi \in P(n,k)$, we denote the subpermutation $\phi(j_1), \phi(j_2), \dots, \phi(j_l)$ of ϕ as $\phi(j_1, j_2, \dots, j_l)$, and the subpermutation of ϕ obtained by removing $\phi(j_1), \phi(j_2), \dots, \phi(j_l)$ as $\tilde{\phi}(j_1, j_2, \dots, j_l)$. Let $P(n, \tilde{\phi}(j_1, j_2, \dots, j_l))$ be the set of $(n-l)$ -permutations of the set $\{a \mid 1 \leq a \leq n \wedge \forall 1 \leq i \leq l, a \neq \phi(j_i)\}$.

From Theorem 1:

$$\begin{aligned} N(k) &= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{i=1}^{n!} \prod_{h=1}^l A_{[j_h, \phi_i(j_h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,n)} \prod_{h=1}^l A_{[j_h, \phi(j_h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi(j_1, j_2, \dots, j_l) \in P(n,l)} \sum_{\tilde{\phi}(j_1, j_2, \dots, j_l) \in P(n, \tilde{\phi}(j_1, j_2, \dots, j_l))} \prod_{h=1}^l A_{[j_h, \phi(j_h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k (n-l)! g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}. \end{aligned} \quad (8)$$

It is evident that if we can compute $\sum_{1 \leq j_h \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}$ in fixed-parameter tractable (FPT) time, then $N(k)$ can be computed in FPT time as well.

We denote $\Pi(n, k)$ as the set of all unordered k -partitions of the set $\{1, 2, \dots, n\}$. That is, if $\{J_1, J_2, \dots, J_k\} \in \Pi(n, k)$, then $J_1 \cup J_2 \cup \dots \cup J_k = \{1, 2, \dots, n\}$, $\forall 1 \leq i_1 \neq i_2 \leq l$, $J_{i_1} \cap J_{i_2} = \emptyset$, and $\forall 1 \leq i \leq l$, $J_i \neq \emptyset$. Furthermore, we denote $S(n, k) = |\Pi(n, k)|$ (the Stirling numbers of the second kind) and $B_n = \sum_{k=1}^n S(n, k)$ (the n -th Bell number).

From the literature, we have:

Lemma 5[3]. $B_n = n^{O(n)}$.

Lemma 6. $\forall k > 1$, there exist functions $f_1^k, f_2^k, \dots, f_k^k$ defined on $\Pi(k, 1), \Pi(k, 2), \dots, \Pi(k, k)$ respectively, such that

$$\sum_{\sigma \in P(n, k)} \prod_{i=1}^k x_{i\sigma(i)} = \sum_{l=1}^k \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k, l)} f_l^k(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}.$$

Proof. We prove this lemma by mathematical induction. The base case is established by Lemma 2, and we proceed to the inductive step:

The inductive hypothesis is

$$\sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} = \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}. \quad (9)$$

Clearly,

$$\begin{aligned} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{\substack{\sigma(k)=1, \sigma(2), \dots, \sigma(k-1) \in P(n, k-1) \\ \sigma(1), \sigma(2), \dots, \sigma(k-1) \neq \sigma(k)}} \prod_{i=1}^k x_{i\sigma(i)} \\ &= \sum_{\sigma(k)=1}^n \sum_{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1)} \prod_{i=1}^k x_{i\sigma(i)} \\ &\quad - \sum_{j=1}^{k-1} \sum_{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1)} x_{k\sigma(j)} \prod_{i=1}^{k-1} x_{i\sigma(i)} \\ &= \sum_{h_k=1}^n x_{kh_k} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} - \sum_{j=1}^{k-1} x_{k\sigma(j)} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)}. \end{aligned} \quad (10)$$

From the inductive hypothesis:

$$\begin{aligned} \sum_{h_k=1}^n x_{kh_k} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{h_k=1}^n x_{kh_k} \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\ &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l, j_{l+1} \leq n} x_{kj_{l+1}} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}. \end{aligned} \quad (11)$$

Moreover,

$$\begin{aligned} \sum_{j=1}^{k-1} x_{k\sigma(j)} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{j=1}^{k-1} x_{k\sigma(j)} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\ &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{i=1}^l |J_i| x_{k\sigma(j_i)} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}. \end{aligned} \quad (12)$$

Substituting (11) and (12) into (10), we obtain:

$$\begin{aligned} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l, j_{l+1} \leq n} x_{kj_{l+1}} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\ &\quad - \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{i=1}^l |J_i| x_{k\sigma(j_i)} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}. \end{aligned} \quad (13)$$

For any partition $\{J_1, J_2, \dots, J_l\} \in \Pi(k, l)$ with $k \in J_l$, we define:

$$f_l^k(\{J_1, J_2, \dots, J_l\}) = \begin{cases} f_{l-1}^{k-1}(\{J_1, J_2, \dots, J_{l-1}\}), & |J_l| = 1, \\ -(l-1)f_{l-1}^{k-1}(\{J_1, J_2, \dots, J_{l-1}, J_l \setminus \{k\}\}), & |J_l| > 1, \end{cases}$$

This completes the proof of the lemma.

Clearly, f_l^k can be computed in $k^{O(k)}$ time using the recursive formula in the proof (this requires evaluating all $\sum_{l \leq m \leq k} S(m, l) = \sum_{m=1}^k B_m$ partitions in $\Pi(m, l)$ for all $l \leq m \leq k$).

Lemma 7.

$$N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k (n-l)! g_k'(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{q=1}^k \sum_{\{P_1, P_2, \dots, P_q\} \in \Pi(k, q)} f_q^k(\{P_1, P_2, \dots, P_q\}) \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}.$$

Proof. This follows immediately by substituting Lemma 6 into Theorem 2 and renaming duplicate symbols.

Theorem 3.

$$N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{l=1}^k (n-l)! g_k'(l) \sum_{q=1}^k \sum_{\{P_1, P_2, \dots, P_q\} \in \Pi(k, q)} f_q^k(\{P_1, P_2, \dots, P_q\}) \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}.$$

Proof. This follows immediately from Lemma 7 by interchanging the order of summation.

Theorem 4. The expression $\sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}$ can be computed in $O(n^2)$ time.

Proof.

$$\begin{aligned} \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]} &= \prod_{h=1}^q \sum_{1 \leq p_h \leq n} \prod_{i \in P_h} \sum_{1 \leq j_i \leq n} A_{[j_i, p_h]} \\ &= \prod_{h=1}^q \sum_{1 \leq p_h \leq n} d_{p_h}^{|P_h|}. \end{aligned} \quad (14)$$

Computing $\prod_{h=1}^q \sum_{1 \leq p_h \leq n} d_{p_h}^{|P_h|}$ only requires calculating $\sum_{1 \leq p_1 \leq n} d_{p_1}^{|P_1|}, \sum_{1 \leq p_2 \leq n} d_{p_2}^{|P_2|}, \dots, \sum_{1 \leq p_q \leq n} d_{p_q}^{|P_q|}$, followed by $(q-1)$ multiplications. The total time complexity is $O(n^2)$.

Theorem 5. $N(k)$ can be computed in fixed-parameter tractable (FPT) time.

Proof. By Lemma 5, $\sum_{l=1}^k S(k, l) = k^{O(k)}$. The formula in Theorem 3 requires evaluating a total of $\sum_{l=1}^k \sum_{q=1}^k B_q$ expressions of the form

$$\sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]},$$

with a time complexity of $k^{O(k)}O(n^2)$. Additionally, the time complexity of deriving the expression in Theorem 3 is $k^{O(k)}$. Thus, $N(k)$ is fixed-parameter tractable (FPT).

It follows directly from Theorem 5 that:

Theorem 6. $\#W[1] = \text{FPT}$.

Theorem 7. The Counting Exponential Time Hypothesis ($\#ETH$) does not hold.

Theorem 8. The Exponential Time Hypothesis (ETH) does not hold.

3 Conclusion

As a classic problem in the interdisciplinary field of counting complexity and parameterized complexity, the #k-matching problem has long been proven by the academic community to be a canonical #W[1]-hard problem, serving as a core benchmark for verifying the existence of parameterized efficient algorithms for such counting problems. We conduct algorithm design and derivation via algebraic methods, where the #k-matching problem refers to the counting problem of enumerating the number of edge-disjoint k-matchings in an undirected graph. Ultimately, we successfully present an $f(k)n^{O(1)}$ -time algorithm for the #k-matching problem, whose complexity fully adheres to the strict definition of fixed-parameter tractability (FPT). This research result directly demonstrates that the long-standing hypotheses of $\#W[1] \neq \text{FPT}$ and $W[1] \neq \text{FPT}$, which have perplexed the field of theoretical computer science for decades, do not hold. Consequently, the Counting Exponential Time Hypothesis ($\#ETH$) and the Exponential Time Hypothesis (ETH), as their theoretical cornerstones, are also falsified.

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